

Artificial Intelligence

1. Linear Regression:

Predicts a dependent variable using one independent variable.

Equation:

$$Y = B_0 + B_1 X$$

Y = Output

X = Input

B_0 = Intercept

B_1 = slope (how much Y changes per unit change in X).

→ Mean of X and Y
 $\bar{X}$ & $\bar{Y}$

→ Compute slope (m) using formula:

$$m = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

2. Multiple Linear Regression:

Predicts a dependant variable using two or more independent variables

Equation:

$$Y = B_0 + B_1 X_1 + B_2 X_2 + \dots + B_n X_n$$

$X_1, X_2, \dots$: Multiple inputs

$B_1, B_2, \dots$: slopes for each input

Normal Equation:

$$\Theta = (X^T X)^{-1} X^T Y$$

X^T = Transpose of X .

$(X^T X)^{-1}$ = Inverse of matrix.

Step by step INVERSE of a matrix:

$$A = \begin{bmatrix} 5 & 15 & 20 \\ 15 & 55 & 70 \\ 20 & 70 & 100 \end{bmatrix} \Rightarrow \text{If } \det(A) \neq 0 \text{ then the inverse exists.}$$

Compute the minors of the matrix.

Examples: $\begin{bmatrix} 55 & 70 \\ 70 & 100 \end{bmatrix}$

$$\text{Minors}(A) = \begin{bmatrix} 600 & 100 & -50 \\ 100 & 100 & 50 \\ -50 & 50 & 50 \end{bmatrix}$$

$$\begin{bmatrix} 15 & 70 \\ 20 & 100 \end{bmatrix}$$

$$\text{Cofactor}(A) = \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \times \begin{matrix} \text{Minors} \\ (A) \end{matrix}$$

$$\text{Cofactor}(A) = \begin{bmatrix} 600 & -100 & +50 \\ -100 & 100 & -50 \\ -50 & -50 & 50 \end{bmatrix}$$

$$\text{Adjugate}(A) = \text{Cofactor}(A)^T = \begin{bmatrix} 600 & -100 & -50 \\ -100 & 100 & -50 \\ -50 & -50 & 50 \end{bmatrix}$$

- Divide by the Determinant

$$\frac{1}{\det(A)} = \frac{1}{500}$$

Now then $A^{-1} = \begin{bmatrix} 6/5 & -1/5 & -1/10 \\ -1/5 & 1/5 & -1/10 \\ -1/10 & -1/10 & 1/10 \end{bmatrix}$

Extra info: During training, we use known X and Y to estimate the β coefficients. Once the model is trained, we can input new X values to predict unknown Y values. So, in training phase, Y is known, but in prediction, Y is what we're trying to find.